Theoretical Model of the Minimum Cumulative Detection Count for a Photon-Counting LiDAR

Zhiyu Zhang, Kanbu Tan, Xinggui Wei, Zheng Zhao, Xinyuan Liu, Xiangcheng Chen, Guangyao Dai, and Songhua Wu

***Abstract*—Photon-counting (PC) light detection and ranging (LiDAR), characterizing single-photon sensitivity and precise timing, plays a critical role in long-range detection, 3-D imaging, and spaceborne missions such as The Ice, Cloud, and land Elevation Satellite-2 (ICESat-2). However, its performance is constrained by the use of a fixed number of accumulated detections, causing inefficient resource allocation in a complex environment. Specifically, under high photon contrast, it leads to redundant time consumption or degraded spatial resolutions, whereas under low photon contrast, it cannot attain statistically sufficient photon events (PEs) for signal discrimination. Moreover, a theoretical foundation for the minimum cumulative detection count remains underdeveloped. To address this issue, this article develops a statistical model to determine the minimum detection count under various photon contrast conditions. The model is derived based on binomial and normal distributions, and further extended to multichannel detection systems and complex scenarios involving photon contrast constraints. Validation via Monte Carlo simulations and experiments shows close agreement with theory, with absolute errors within 2.4% (simulation) and 1.79% (experiment), and a maximum deviation below 5%. The model reduces required detections by up to 50% under high contrast while maintaining over 90% accuracy. Furthermore, when applied to a scanning imaging LiDAR system, the method improves the structural similarity index measure (SSIM) by 23% under a photon contrast constraint $Q = 2$, compared to conventional equal-interval scanning within equal acquisition time. This study provides a scalable framework for PC LiDAR signal processing, with strong potential for enhancing spaceborne LiDAR data processing in complex environments.**

***Index Terms*—Light detection and ranging (LiDAR) remote sensing, multichannel detectors, photon contrast, photon-counting (PC) LiDAR, scanning imaging, the minimum cumulative detection count.**

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I. Introduction

PHOTON-COUNTING (PC) light detection and ranging (LiDAR), characterized by single-photon sensitivity and precise timing capability, enables the detection of extremely weak signals beyond the reach of conventional detectors [1], especially in long-range or highly attenuated environments. A prominent example is the ICESat-2 mission, launched in 2018, whose onboard PC LiDAR has made groundbreaking contributions to ice sheet monitoring, sea ice thickness estimation, and vegetation canopy mapping [2], [3]. In parallel, PC LiDAR-based 3-D imaging has emerged as a leading active sensing technique in recent years [4], [5], demonstrating significant application potential in low-illumination imaging [6], [7], [8], long-range detection [9], [10], [11], and underwater target identification [12].

However, this high detection sensitivity is achieved at the expense of intensity discrimination in a single measurement [13], [14]. As a result, PC LiDAR must accumulate multiple detections to estimate target reflectivity through the statistical distribution of signal photon events (PEs) [15], [16]. This cumulative detection strategy is particularly exemplified in LiDAR remote sensing applications, such as the advanced topographic laser altimeter system (ATLAS) onboard ICESat-2, which has marked a new era in PC altimetry. The data processing algorithms for ATLAS are considerably more complex than those of conventional linear-mode LiDAR systems. ATLAS adopts target-specific accumulation lengths to construct PE histograms for signal extraction and geophysical parameter retrieval. For instance, the strong beam accumulates 400 shots (at 25 Hz) to produce atmospheric profiles [17], while land surface signals are extracted using only 50 accumulated shots (at 200 Hz) [18]. In ocean profiling, the received backscattered signal is extremely weak—typically on the order of 10^{-3} to 10^{-2} photon counts per meter depth per laser pulse in clear waters—implying that roughly 100 to 1000 shots are needed on average to register a single PE [19]. Notably, the accumulation counts used across these scenarios remain largely empirically determined, underscoring the need for a rigorous theoretical basis for selecting the minimum cumulative detection count, particularly for future spaceborne missions operating under dynamically varying photon fluxes.

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The choice of cumulative detection count thus critically influences signal identification performance, especially under conditions of relative sensor-target motion or fluctuating signal and noise levels. Insufficient detections can lead to inaccurate statistical representation of the signal PE distribution [20], whereas excessive accumulations may impair dynamic target tracking capability or degrade spatial resolution [21]. Currently, the selection of the cumulative detection count in PC LiDAR systems relies predominantly on empirical knowledge [22], [23]. It is therefore essential to determine an appropriate accumulation number that balances performance and efficiency—specifically, the minimum cumulative detection count required for reliable signal discrimination.

Existing studies on the minimum cumulative detection count exhibit several limitations. Niclass et al. [24] reconstruct the depth information of arbitrary 3-D objects using the time-of-flight (TOF) of photons and established a theoretical mapping relationship between the cumulative detection count and ranging accuracy. They concluded that achieving millimeter-level ranging accuracy requires more than 104 PEs. Kirmani et al. [25] proposed the first-photon imaging (FPI) method, which only requires one photon to be detected per target unit to retrieve the radiative and geometric information of the target, demonstrating astonishing imaging efficiency. However, this method is relatively sensitive to background noise, and its robustness under scenarios with varying photon contrast leaves room for further enhancement. Hou et al. [26] developed a performance boundary model applicable to different photon fluxes to establish the relationship between the cumulative detection count and waveform restoration accuracy in various application scenarios. Nevertheless, this method lacks sensitivity analysis on the discrimination between signal and noise. Chen et al. [27] proposed a model relating physical parameters to minimum ranging time but did not provide an analytical expression for the detection count under false alarm rate constraints. Overall, a systematic and quantitative framework for determining the minimum cumulative detection count in PC LiDAR is still lacking.

In this article, we propose a theoretical model for determining the minimum cumulative detection count required to discriminate signal from noise in PC LiDAR. The model is first derived based on binomial statistics under a false alarm rate constraint and then approximated using the normal distribution to yield an analytical expression suitable for engineering applications. We validate the model through both Monte Carlo simulations and laboratory experiments. The framework is extended to multichannel single-photon detector configurations, and the impacts of photon contrast, time bin width, and range gate length are thoroughly analyzed. Finally, the model is applied in a scanning imaging LiDAR system and benchmarked against conventional equal-interval scanning, confirming its validity and robustness.

II. Theoretical Model of the Minimum Detection Count for a PC LiDAR

During a single detection of PC LiDAR, it is difficult to discriminate whether a PE originates from a signal or noise photon. Consequently, multiple detections are accumulated to form a histogram, which is then used for analysis. Gardner defined photon contrast as the ratio of signal PE counts to noise PE counts, as shown in (1). This metric replaces the signal-to-noise ratio for characterizing signal quality in PC LiDAR [28]

$$D = \frac{n_{\mathrm{PEn}} + n_{\mathrm{PEs}}}{n_{\mathrm{PEn}}} \approx \frac{P_{\mathrm{n}} + P_{\mathrm{s}}}{P_{\mathrm{n}}}. \tag{1}$$

In the equation, n_{PEn} denotes the average noise photon number within the signal region, while n_{PEs} denotes the average signal photon number that reaches the detector. P_{s} and P_{n} denote the detection probabilities of signal and noise PEs, respectively. Although the expected value of photon contrast, by definition, remains unchanged with increased detections, practical signal-noise discrimination becomes difficult when the detection count is low. Currently, the selection of the cumulative detection count in most signal processing workflows relies on empirical knowledge due to a lack of a theoretical model. In this section, based on the probability distributions of signal PEs and noise PEs, we propose a minimum cumulative detection count model. This model explains why the increasing of accumulated detection count enhances the ability to discriminate signal from noise, then it provides an analytical solution to the minimum cumulative detection count for a PC LiDAR.

A. Statistical Model of Signal and Noise PEs

In practice, the number of PEs received within a single time bin over a period is typically used to discriminate signal from noise, where the expected signal PE counts should significantly exceed the expected noise PE counts. Therefore, this article first establishes models for the signal and noise PE counts received over N-time of detections.

According to Goodman's theory, the probability of a single-photon detector observing K PEs within a single time bin follows the Poisson distribution when the probability of observing a PE in that bin approaches zero [29], [30], [31]. Given that single-photon detectors have a dead time td, if the average number of received echo photons is denoted by $S(\mathrm{t})$ and the width of a time bin is τ, the detection probability of signal PEs within a specific time bin can be expressed as follows [32]:

$$\begin{aligned} &P_{\mathrm{s}}(t, t+\tau; K>0) \\ &= \exp\left(-\int_{t-t_{\mathrm{d}}}^{t} S(t)\,\mathrm{d}t\right) \times \left[1-\exp\left(-\int_{t}^{t+\tau} S(t)\,\mathrm{d}t\right)\right] \end{aligned} \tag{2}$$

where the first term represents the probability that no PE is detected within the interval $[t - t_d, t]$, while the second term represents the probability of detecting a PE within $[t, t + \tau]$. Meanwhile, the detection probability of noise PEs within a single bin can be expressed as follows:

$$P_{\mathrm{n}}(t, t+\tau; K>0) = \exp(-f_{\mathrm{n}} t_{\mathrm{d}}) \times \left[1-\exp(-f_{\mathrm{n}}\tau)\right]. \tag{3}$$

Under this condition, the probability function for the received signal and noise PE count over N detections can be further derived. In scenarios characterized by steady background illumination and the use of laser pulses with high

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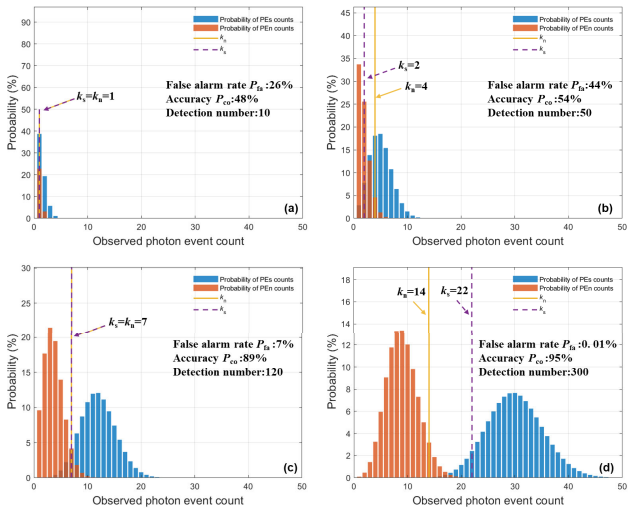


Fig. 1. Probability function of signal and noise PE counts under different numbers of detections. (a)–(d) Correspond to the detection count of 10, 50, 120, and 300, respectively, under the condition that the detection probabilities of signal and noise PEs remain constant as P_s = 10% and P_n = 3%. The blue and orange histograms represent the probability function of the signal and noise PE counts, respectively.

stability, the probability of the detector capturing a PE in each discrete trial remains invariant. Under these conditions, the probability density functions (PDFs) of the signal and noise PEs can be modeled as binomial distributions, i.e., $N_{PEs} \sim B(N, P_s)$, $N_{PEn} \sim B(N, P_n)$ [33].

Fig. 1 shows the probability function of the signal and noise PE counts under different detection counts, when the detection probabilities of the signal (blue) and noise (orange) PEs are fixed to 10% and 3%, respectively. The abscissa represents the observed PE counts, and the ordinate represents their probability. The yellow solid line indicates the 95th percentile (k_n) of the noise PE count, while the purple dashed line indicates the fifth percentile (k_s) of the signal PE count. Since the binomial distribution is discrete, only approximate values can be obtained for the k_s and k_n. Here, k_s is the maximum value of n_{PEs} satisfying (4), and k_n is the minimum value of n_{PEn} satisfying (5)

$$\sum_{n=0}^{n_{PEs}} C_N^n P_s^n (1-P_s)^{N-n} \le 5\% \tag{4}$$

$$\sum_{n=0}^{n_{PEn}} C_N^n P_n^n (1-P_n)^{N-n} \ge 95\%. \tag{5}$$

As shown in Fig. 1, the probability function of the signal and noise PE count progressively separates as the accumulated detections increase, and a threshold can be determined to discriminate signal from noise when the overlapping area between the two distributions becomes sufficiently small. Taking Fig. 1(b) as an example, at 50 detections, using the k_s as the threshold would result in an unacceptable probability of noise PEs being misclassified as signal PEs. As shown in Fig. 1(d), when the detection count reaches 300, the two distributions separate further, and their overlapping probability decreases substantially. If k_s is still used as the threshold, the probability of misclassification will decrease to about 0.1‰. Thus, when the k_s is larger than the k_n, there always exists a suitable accumulated detection counts N_{min} that enables discrimination between signal and noise under given false alarm rate constraints. The utilization of the 5th and 95th percentiles as decision thresholds is a standard practice in statistical inference, established within the framework of hypothesis testing and the construction of confidence intervals [34], [35]. In this study, the signal-noise discrimination criterion is defined such that the fifth percentile of the signal PE distribution must exceed the 95th percentile of the noise PE distribution. Under this condition, we define the minimum cumulative detection count N_{min} as the value for which $k_n < k_s$ first holds true, and the N_{min} can be expressed as follows:

$$N_{min} = \min\left\{N \in Z^+ : k_n(N) < k_s(N)\right\} \tag{6}$$

where Z^+ denotes the set of positive integers.

Building upon this foundation, we refer to Degnan's definitions of accuracy rate and false alarm rate for distinguishing signal from noise [36]. Furthermore, in calculating the accuracy rate P_{co}, we have also incorporated consideration of noise PEs. The accuracy of signal discrimination P_{co} at N_{min} detection, defined as the probability where the signal PE count exceeds the threshold k_s while the noise PE count remains below k_s, which can be derived as follows:

$$P_{co} = \sum_{n_{PEn}=0}^{k_s} C_N^{n_{PEn}} P_n^{n_{PEn}} (1-P_n)^{N-n_{PEn}} \times \sum_{n_{PEs}=k_s}^{\infty} C_N^{n_{PEs}} P_s^{n_{PEs}} (1-P_s)^{N-n_{PEs}}. \tag{7}$$

Meanwhile, the false alarm rate P_{fa} can be expressed as the probability that the noise PE count exceeds the threshold k_s

$$P_{fa} = \sum_{n_{PEn}=k_s}^{\infty} C_N^{n_{PEn}} P_n^{n_{PEn}} (1-P_n)^{N-n_{PEn}}. \tag{8}$$

Considering the condition in Fig. 1(b), at only 50 detections, using k_s as the threshold yields a discrimination accuracy of merely 54% with a false alarm rate up to 44%. When the detection count increases to 120, the k_s coincides exactly with k_n, the accuracy rises to 89%, while the false alarm rate drops to 7%. At 300 detections [Fig. 1(d)], the overlapping area between the distribution of the noise and signal PE count further reduces, and the accuracy reaches 95% with a false alarm rate of only 0.1‰. This phenomenon demonstrates that while increasing the detection count cannot enhance photon contrast, it improves the accuracy of signal discrimination and reduces the false alarm rate. Fig. 2 illustrates the N_{min} required for discriminating signal photons, along with its accuracy and false alarm rate, under varying average signal photon number and background noise. In Fig. 2(a), the contour lines denote the N_{min}, calculated using (6). In Fig. 2(b) and (c), the contour lines represent the discrimination accuracy and false alarm rate, respectively.

As shown in Fig. 2, the N_{min} generally increases as the photon contrast decreases. When N_{min} exceeds 200, the discrimination accuracy and the false alarm rate stabilize

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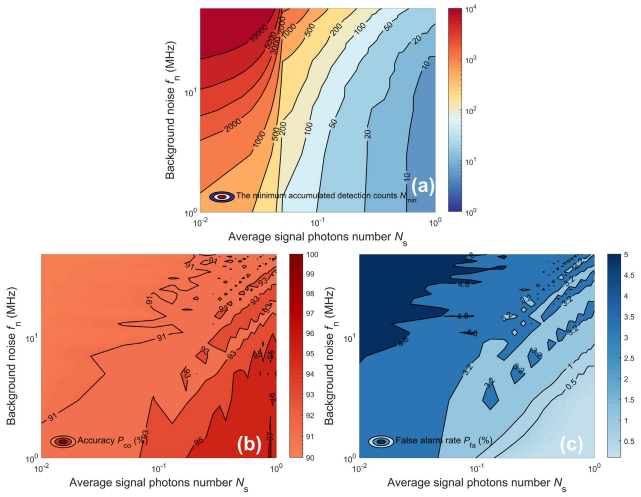


Fig. 2. (a) Contour map of the minimum cumulative detection count $N_{\min}$ required for signal discrimination under varying signal and background noise conditions. (b) and (c) Contour maps of the discrimination accuracy and false alarm rate, respectively, corresponding to results using k_s as the threshold at $N_{\min}$ detections.

respectively at 90% and 5%. Conversely, when the $N_{\min}$ is less than 20, the accuracy rate reaches up to 97% while the false alarm rate reduces to 0.5%. This result arises from the discrete characteristic of the binomial distribution. Under low detection count, the cumulative probabilities associated with the k_s and k_n percentiles exhibit deviation from their theoretical values of 5% and 95%. As delineated by (7) and (8), these conditions yield a discrimination accuracy that exceeds 95% and a false alarm rate below 5%. For larger cumulative detections, the deviations become negligible. Consequently, the accuracy approaches 90% while the false alarm rate converges to 5%.

B. Analytical Model Based on Normal Distribution Approximation

In Section II-A, a theoretical model of minimum cumulative detection count is derived. However, the discrete characteristic of the binomial distribution complicates obtaining an analytical solution, making it inconvenient for engineering applications. Moreover, when the photon contrast is low, the minimum cumulative detection count, herein designated $N_{\min}$, will experience a significant increase, rendering the solution of $N_{\min}$ inefficient. Given that this issue predominantly arises for large $N_{\min}$, we can reasonably approximate the binomial distribution with a normal distribution to compute $N_{\min}$.

Under this condition, the pdf of the detected signal and noise PE counts can be approximated as $N_{\mathrm{PEs}} \sim \mathrm{N}(\mu_s, \sigma_s^2)$, $N_{\mathrm{PE}n} \sim \mathrm{N}(\mu_n, \sigma_n^2)$. The parameters of the normal distributions relate to the binomial distribution parameters as follows [37]:

$$\mu_x = NP_x, \sigma_x^2 = NP_x(1-P_x) x \in \{s, n\} \tag{9}$$

where μ_x and σ_x^2 denote the mean and variance of the normal distribution, and P_x represents the detection probability in (2) and (3). The subscript $x \in \{s, n\}$ corresponds to signal and noise, respectively. And the k_s and k_n can be expressed

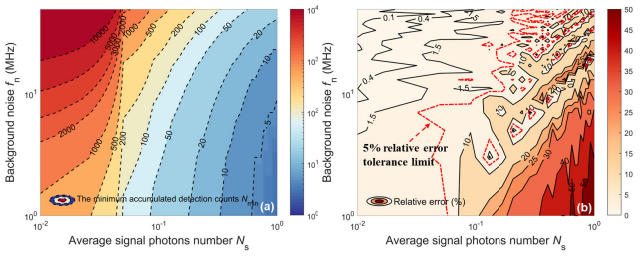


Fig. 3. (a) Contour map of the theoretical minimum cumulative detection count $N_{\min}$ calculated using normal distribution. (b) Contour map of the relative error between $N_{\min}$ computed via normal and binomial distributions. Contour labels indicate percentage relative error, which generally increases as $N_{\min}$ decreases. The red dashed contour lines indicate the tolerance boundary for a 5% relative error in high photon contrast regions.

as $\mu s - 1.65\sigma_s$ and $\mu_n + 1.65\sigma_n$, respectively. This allows the derivation of a closed-form expression for $N_{\min}$ under the normal approximation as follows:

$$\begin{aligned} &N_{\min}P_s - 1.65\sqrt{N_{\min}P_s(1-P_s)} \\ &\geq N_{\min}P_n + 1.65\sqrt{N_{\min}P_n(1-P_n)}. \end{aligned} \tag{10}$$

Further arrangement yields the analytical expression of $N_{\min}$ under the normal distribution approximation as follows:

$$\begin{aligned} N_{\min} \geq\ & \frac{2.72(P_s - P_s^2) + 2.72(P_n - P_n^2)}{(P_s - P_n)^2} \\ &+ \frac{5.44\sqrt{P_nP_s(1-P_n)(1-P_s)}}{(P_s - P_n)^2} \end{aligned} \quad (P_s > P_n). \tag{11}$$

Accordingly, the discrimination accuracy at $N_{\min}$ detections is given by the following equation:

$$P_{co} = \frac{\sqrt{\sigma_s\sigma_n}}{4}\mathrm{erf}(k_s) \cdot \mathrm{erfc}(k_s) \tag{12}$$

where $\mathrm{erf}(k_s)$ denotes the Gaussian error function and $\mathrm{erfc}(k_s)$ is the complementary Gaussian error function. These functions are defined as follows:

$$\begin{cases} \mathrm{erf}(k_s) = \dfrac{\sqrt{2}}{\sqrt{\pi}\sigma_n}\displaystyle\int_0^{k_s} \exp\left(-\frac{(n_{\mathrm{PEn}} - \mu_n)^2}{2\sigma_n^2}\right) dn_{\mathrm{PEn}} \\ \mathrm{erfc}(k_s) = \dfrac{\sqrt{2}}{\sqrt{\pi}\sigma_s}\displaystyle\int_{k_s}^{+\infty} \exp\left(-\frac{(n_{\mathrm{PEs}} - \mu_s)^2}{2\sigma_s^2}\right) dn_{\mathrm{PEs}}. \end{cases} \tag{13}$$

The false alarm rate can be expressed as follows:

$$P_{fa} = 1 - \frac{1}{2}\mathrm{erf}(k_s). \tag{14}$$

Since the normal approximation requires the cumulative detection count N to approach infinity, errors may arise when $N_{\min}$ is small. Fig. 3 shows $N_{\min}$ calculated using the normal approximation and its relative error under varying signal and noise conditions. Fig. 3(a) demonstrates the contour map of $N_{\min}$ under the normal approximation. Fig. 3(b) presents the relative error between $N_{\min}$ computed via normal and binomial distributions. Compared with the results calculated by (6), the normal approximation yields systematically lower values. The relative error concentrates primarily in regions with strong photon contrast, while in regions with low average signal photon number (where $N_{\min}$ exceeds 100), the relative error remains below 5%.

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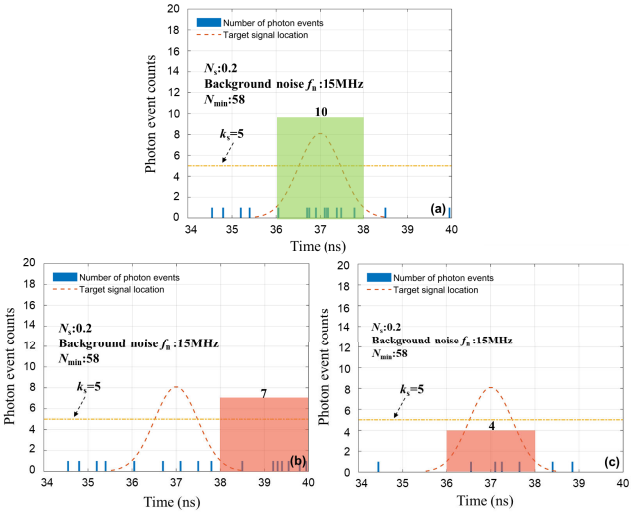


Fig. 4. Schematics of signal discrimination results based on PE count. (a) Successful signal identification, where the PE count of a time bin (marked with the green rectangle) is higher than k_s and its position matches the target signal. (b) and (c) Two cases of failed discrimination. In (b), although the PE count exceeds the threshold, it locates outside the signal region. In (c), no time bin reaches the threshold.

For regions with favorable photon contrast, although the relative error associated with the normal distribution calculation is relatively large, the N_{min} value itself is typically small, which means the consistency can be improved by simply increasing a few detections. It is also feasible to directly employ the binomial distribution for computation with little computational efficiency loss. However, when photon contrast is suboptimal, the N_{min} will increase exponentially with the rising noise rate. In such cases, utilizing the analytical solution approximated by the normal distribution can significantly enhance computational efficiency while maintaining the consistency. Consequently, this article recommends using (11) to calculate N_{min} when the number of signal photons falls within 0.01 to 0.1 and the noise rate is between 1 and 50 MHz. When the number of signal photons exceeds 0.1, it is advised to compute N_{min} directly using (6).

III. Verification

This study validates the theoretical model using both Monte Carlo simulations and a PC LiDAR system. In practice, the most widely adopted signal discrimination method compares the maximum PE count in the accumulated histogram against a threshold to determine whether it contains signal. Our analysis of simulation and experimental results employs a similar criterion. Fig. 4 illustrates this criterion in detail. The abscissa represents the time tag of each PE, while the ordinate indicates the corresponding PE count. The blue bars show the PE count of each physical time bin under N_{min} detections. The yellow dashed line indicates the theoretical discrimination threshold k_s, and the red dashed line delineates the envelope of the reflected signal waveform.

During the signal discrimination procedure, first, PEs are divided into several "virtual" time bins based on the signal pulsewidth ($\pm 2\sigma_s$), and the time bin containing the highest number of PEs is identified—for example, the regions marked by green and red rectangles in Fig. 4. Next, we determine whether the PE count in this time bin exceeds the discrimination threshold k_s. If this condition is met, we further check whether the time slice falls within the time interval corresponding to the target. If it does, the signal is considered successfully identified [Fig. 4(a)]; otherwise, in the cases shown in Fig. 4(b) and (c), the identifications are deemed failure.

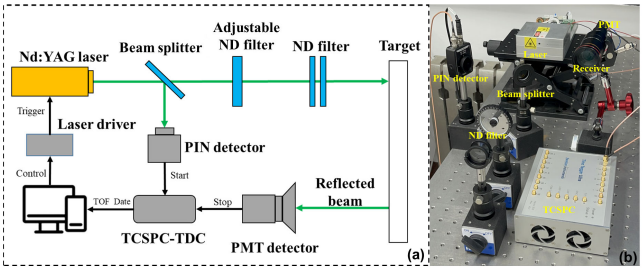


Fig. 5. (a) Schematic of the PC LiDAR system. (b) Photograph of the experimental system.

A. Monte Carlo Simulation Procedure

Leveraging our team's pre-established PMT detector response model and Monte Carlo simulation framework [38], [39], this work simulates PMT-based signal detection. The Monte Carlo simulation procedure for a single detection is as follows.

1) Generate Number of Signal and Noise Photons: The number of signal photons N_s and noise photons N_n follow Poisson distributions, with $n_s \sim P(= N_s)$ and $n_n \sim P(\lambda = N_n)$, respectively.

2) Assign Photon Arrival Times: The time tags of signal photons are generated from a normal distribution, and those of noise photons from a uniform distribution.

3) Simulate PMT's Output Current: Process signal and noise photons through the PMT response model. Add output currents to generate the analog waveform.

4) Generate PE: Set an output current threshold. Record only the first PE exceeding this threshold. Discard all subsequent events until the current drops below the threshold. This forms the final PE dataset.

5) Form Cumulative Histogram: Randomly select N_{min} detections in the PE dataset and form a cumulative histogram.

B. Experiment Setup

Fig. 5(a) presents a schematic of the PC LiDAR system, while Fig. 5(b) shows its physical implementation. The experimental setup utilizes a solid-state Nd:YAG laser system, where the 1064-nm fundamental output is frequency-doubled to 532 nm. The system generates laser pulses at a 50-kHz repetition rate with a pulsewidth of 1.16 ns [full width at half maximum (FWHM)]. The selection of a 50-kHz repetition rate and a 1.16-ns pulsewidth (FWHM) is informed by both statistical requirements and established industry benchmarks. Physically, the 20-μs interval between pulses ensures that

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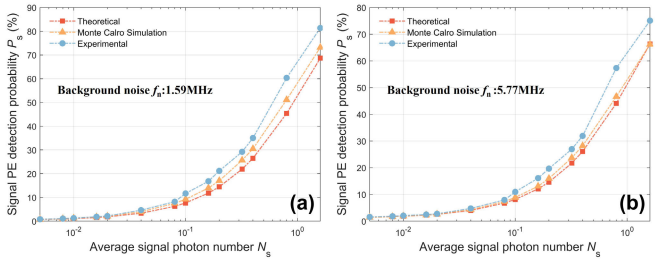


Fig. 6. Theoretical, Monte Carlo simulated, and experimental target detection probabilities under varying background noise. (a) Background noise: 1.59 MHz. (b) Background noise: 5.77 MHz.

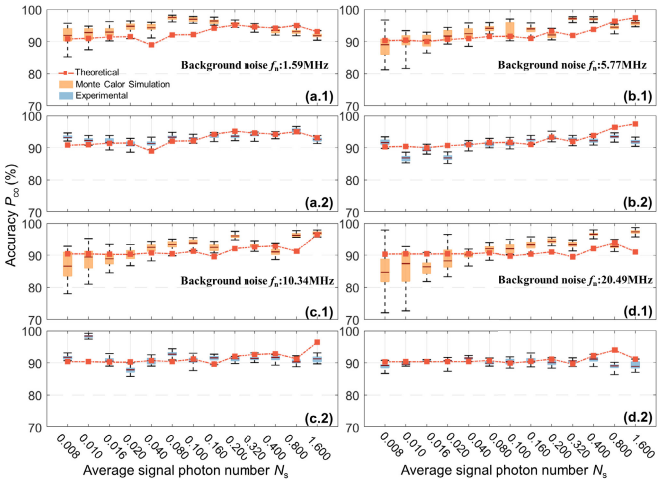


Fig. 7. Theoretical, Monte Carlo simulated, and experimental accuracy following $N_{\min}$ detections under varying background noise. Panels (a.1) and (a.2): Background noise = 1.59 MHz. Panels (b.1) and (b.2): Background noise = 5.77 MHz. Panels (c.1) and (c.2): Background noise = 10.34 MHz. Panels (d.1) and (d.2): Background noise = 20.49 MHz. Boxplots depicting uncertainty components are defined as: median (red line), 25th–75th percentiles (box edges), and extreme data values (whiskers).

detection events remain statistically independent, as this duration significantly exceeds both the pulsewidth and the detector dead time. This independence is essential for the validity of the binomial distribution utilized in our model. Furthermore, these parameters align with the specifications of prominent PC systems—such as Sigma Space's Leafcutter—which typically operate with repetition rates of 10–50 kHz and sub-nanosecond pulse widths to optimize the trade-off between range precision and data throughput [40], [41], [42], [43]. Through a beam splitter, a partial laser beam is directed to a p-i-n photodiode (Thorlabs DET10A2) serving as the timing trigger, while the attenuated beam illuminates the target. Reflected beam is captured by the optical system and focused onto the photocathode of a PMT (Hamamatsu 10682-210). The PMT's output connects to a time-correlated single photon counting (TCSPC) module (Swabian Time Tagger 20 Ultra) for PE timestamping with 8 ps resolution. Attenuation is achieved through an adjustable attenuator with an optical density (OD) range of (OD 0 to 4.0), in conjunction with neutral density filters (OD values of 0.3, 0.5, 1, 2, and 3).

The experiments begin with the attenuation assembly set to maximum attenuation to ensure a low average signal photon number N_s. Subsequently, different N_s levels are implemented by adjusting the attenuation. The background noise in the experimental setup is generated by an adjustable LED light positioned above the optical path. This source provides diffuse illumination that, upon being reflected or scattered by the environment and target, is captured by the receiver.

C. Verification by Simulation and Experimental Results

In practice, the average signal photon number N_s and the background noise f_n are employed as the initial input parameters for calculation. However, due to the dead time effect of the PMT, the detection probability P_s is used to evaluate the actual PE counts in each time bin, which is computed via our PMT detection model, while the simulated and experimental P_s represent the ratio of signal PEs to total detections [38], [39].

Fig. 6 illustrates the comparison results, where the abscissa represents the N_s, while the ordinate denotes detection probability P_s. The red, yellow, and blue lines correspond to theoretical, Monte Carlo simulated, and experimental results, respectively. Comparison of these curves reveals close agreement among theoretical, simulated, and experimental results. As the signal intensity increases, the dead time effect of the single-photon detector becomes more pronounced, introducing measurable deviations between the three methodologies. Despite this divergence, all results remain within 10% error tolerance.

Given the observed discrepancies between experimental P_s values and theoretical predictions, we mitigate potential errors from parameter mismatch by deriving the P_s of simulation and experiment from the statistical value of their own results. These empirical P_s values are then substituted into (11) to calculate the $N_{\min}$ and the corresponding threshold k_s.

To account for the stochastic nature of individual validation trials, the discrimination accuracy P_{co} for each data point is derived from 20 000 statistical trials. These trials are grouped into 20 sets of 1000 trials each to calculate the mean accuracy.

Fig. 7 illustrates the theoretical, simulated, and experimental results under varying signal and noise conditions. The abscissa denotes the average signal photon number N_s, while the ordinate represents the P_{co} at $N_{\min}$ detections. Both the Monte Carlo simulations and the experimental results fluctuate around the theoretical value, with mean absolute errors of 2.4% and 1.79%, respectively. The maximum deviation between experimental and theoretical results remains below 5%. As shown in Fig. 7(a.1) and (a.2), in high photon-contrast regions (N_s = 0.5–1.6), $N_{\min}$ is slightly overestimated, causing the theoretical accuracy P_{co} to increase from 90% to approximately 95%—a trend confirmed by both simulations and experiments. Conversely, in the high-noise environments shown in Fig. 7(b) and (c) (10.34 and 20.49 MHz), the results remain consistently around 90%, further validating the accuracy and reliability of the minimum cumulative detection count model.

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IV. Discussion and Application

A. Minimum Detection Counts for Multichannel Detectors

While single-photon detectors offer high sensitivity for detecting weak signals, they suffer from dead-time-induced saturation under strong signals, limiting the dynamic range of the system. To address this limitation, multichannel detector is adopted to enhance the dynamic range [4], [11], [44], [45], [46], [47]. A representative implementation is the 4 × 4 multianode PMT deployed by the ATLAS onboard ICESat-2 [2].

Consider a n_c-channel PC detector where incident photons are uniformly distributed across channels. The signal detection probability of each channel becomes

$$P_{sc}(t, t+\tau; K>0) = \exp\left(-\int_{t-t_d}^{t} S(t)\,dt/n_c\right) \times \left[1-\exp\left(-\int_{t}^{t+\tau} S(t)\,dt/n_c\right)\right]. \tag{15}$$

Correspondingly, the probability of detecting a noise PE in one channel is

$$P_{nc}(t, t+\tau; K>0) = \exp(-f_n t_d/n_c) \times \left[1-\exp(-f_n \tau/n_c)\right]. \tag{16}$$

If the amplifier circuits and timing systems of each detector channel are independent, and the crosstalk and inconsistencies between channels are ignored, the pdf of signal and noise PE counts detected by the multichannel detector can be expressed as $N_{PEs} \sim N(Nn_cP_{sc}, Nn_cP_{sc} - Nn_cP_{sc}^2)$, $N_{PEn} \sim N(Nn_cP_{nc}, Nn_cP_{nc} - Nn_cP_{nc}^2)$, respectively. The minimum cumulative detection count N_{min} of the multichannel single-photon detector can be expressed as follows:

$$N_{min} \geq \frac{2.72\left(P_{sc}-P_{sc}^2\right)+2.72\left(P_{nc}-P_{nc}^2\right)}{n_c\left(P_{sc}-P_{nc}\right)^2} + \frac{5.44\sqrt{P_{sc}P_{nc}(1-P_{sc})(1-P_{nc})}}{n_c\left(P_{sc}-P_{nc}\right)^2}. \tag{17}$$

Fig. 8 illustrates the relationship between the channel number and the N_{min} for multichannel single-photon detectors. Theoretical calculations assume a background noise rate of 20 MHz, with square, circular, triangular, and diamond markers denoting N_{min} for detectors with 1, 2, 4, and 16 channels, respectively. Under constant photon contrast conditions, the N_{min} decreases as channel count increases. Simulation results confirm that multichannel architectures not only enhance system dynamic range but also reduce detection counts required for signal discrimination. This reduction proves significant at low photon contrast, where additional channels yield substantial efficiency gains. Conversely, at high photon contrasts, the marginal benefit of channel expansion diminishes noticeably.

In practical applications, multichannel single-photon detectors are often affected by the inter-channel crosstalk and nonuniform signal photon distributions across the array. These factors can degrade the performance of PC LiDAR systems and directly impact the calculation of N_{min}. To investigate these effects, we perform Monte Carlo simulations based on a

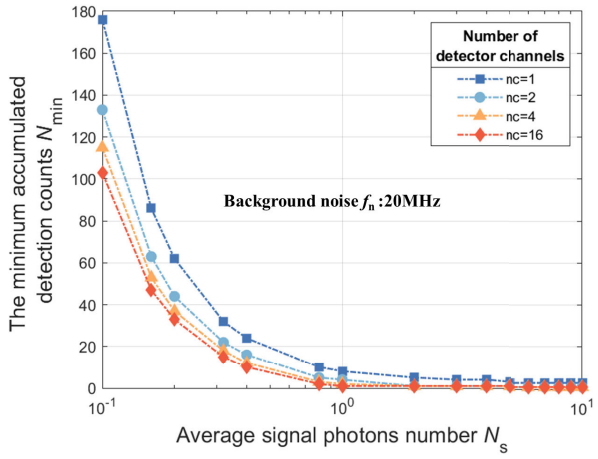


Fig. 8. Minimum detection count N_{min} required for multichannel single-photon detectors. In the figure, curves with rectangular, circular, triangular, and diamond markers correspond to detectors with channel numbers n_c = 1, 2, 4, and 16, respectively. For a given photon contrast, single-photon detectors with more numbers of channels require fewer detections. However, the marginal benefit decreases as the channel number increases.

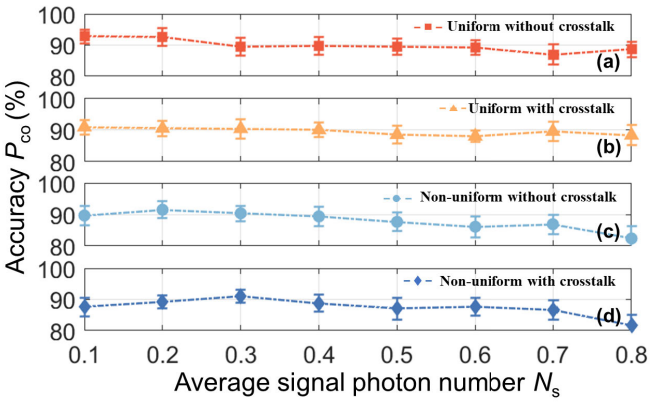


Fig. 9. Discrimination accuracy rate P_{co} at a background noise rate of 20 MHz. (a) Ideal condition: uniform photon distribution without crosstalk. (b) Effect of inter-channel crosstalk. (c) Effect of nonuniform distribution of signal photons. (d) Combined effects of inter-channel crosstalk and nonuniform signal photon distribution.

Hamamatsu 4 × 4 multianode PMT (R7600-300-M16) under a background noise rate of 20 MHz and an average signal photon number N_s ranging from 0.1 to 0.8. The simulation modeled a Gaussian signal distribution across the anodes and introduced adjacent-channel crosstalk with a probability of 0.5%, consistent with Hamamatsu photomultiplier tube specifications.

Fig. 9 presents the simulation results that comprehensively account for the combined effects of nonuniform photon distribution and inter-channel crosstalk in the 4×4 multianode PMT. The horizontal axis denotes the average signal photon number N_s, and the vertical axis represents the discrimination accuracy P_{co}. The curves in Fig. 9 illustrate P_{co} under four conditions: uniform distribution without crosstalk, uniform distribution with crosstalk, nonuniform distribution without crosstalk, and nonuniform with crosstalk. A comparison among these curves reveals that nonuniform signal photon distribution results in a lower P_{co} relative to the uniform case, and this

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discrepancy becomes more pronounced at higher N_s. This behavior arises because, for a fixed N_{min}, channels with weaker illumination receive fewer signal photons while maintaining the noise counts, which lowers the efficiency of reliable signal extraction.

Regarding inter-channel crosstalk, spurious PE PEc increases both signal and noise counts. Under high photon-contrast conditions, this can lead to an overestimation of the signal-to-noise contrast, resulting in an underestimation of N_{min} and a subsequent reduction in P_{co}. However, the presence of detector dead-time creates a complex interplay between crosstalk-induced spurious events and the photon-loss (masking) effect, making the net influence of crosstalk on P_{co} ambiguous. These results confirm that while non-ideal factors influence performance, the minimum cumulative detection model remains an effective framework for signal discrimination.

B. Minimum Detection Counts Under Photon Contrast Constraints

In practical applications, if further processing of signals is required, such as surface signal identification in complex terrains, vegetation canopy detection, or inversion of water optical parameters [49]—merely discriminating whether a time bin contains signal photons may fail to meet the required photon contrast. In Section II-A, this article derived the minimum cumulative detection count N_{min} required to ensure that the signal PE count exceeds that of noise PE with a 90% probability. However, to satisfy the photon contrast required for specific applications, the N_{min} needs to be further increased. Suppose a photon contrast Q is required [$Q < D$, where D denotes the theoretical upper limit defined by (1)]. This problem transforms into ensuring that the signal PE count within a time bin exceeds that of noise PE by a factor of $Q-1$ with over 90% probability. Taking the scenario in Fig. 1 ($P_s = 10\%$, $P_n = 3\%$) as an example, while a satisfactory accuracy in signal discrimination can be achieved when the detection count reaches 300, there remains a small possibility that the signal PE count is comparable to that of noise PE, i.e., a low photon contrast.

Fig. 10 illustrates an extended model for achieving a designed photon contrast, such as $Q = 4.3$, with $\geq 90\%$ probability. In theoretical derivation, when the condition ($n_{PEs} > (Q-1)\, n_{PEn}$) holds, it is often necessary to construct a new joint probability distribution for rigorous analysis, e.g., a stochastic Event $\boldsymbol{A}$ as $\boldsymbol{A} = n_{PEs} - (Q-1)\cdot n_{Pen}$. Since both signal ($n_{PEs}$) and noise ($n_{PEn}$) events follow normal distributions, it follows that Event $\boldsymbol{A}$ also adheres to a normal distribution, $N_A \sim N(\mu_A, \sigma_A^2)$, where $\mu_A = \mu_s - (Q-1)\,\mu_n$ and $\sigma_A^2 = \sigma_s^2 + (Q-1)\sigma_n^2$. If we require $P(A > 0) \geq 90\%$, it can be simplified to require the 10th percentile of $\boldsymbol{A}$ to be larger than 0. The N_{min} can be rigorously derived by substituting (9) to the inequation, i.e., $\mu_A - 1.2816\sigma_A \geq 0$. To facilitate comprehension and application, here we follow the prior work and introduce an equivalent noise PE count N_{VPEn} (the yellow histogram), whose equivalent detection probability P_{vn} is $(Q-1)\cdot P_n$. And the green dashed line still denotes the fifth percentile (k_s) of the signal PE count, while the purple dashed line now indicates the 95th percentile (k_n) of N_{VPEn}. The pdf of N_{VPEn} also satisfies a normal distribution, i.e., $N_{VPEn} \sim N(\mu_{vn}, \sigma_{vn}^2)$, where μ_{vn} and σ_{vn}^2 can be respectively expressed as follows:

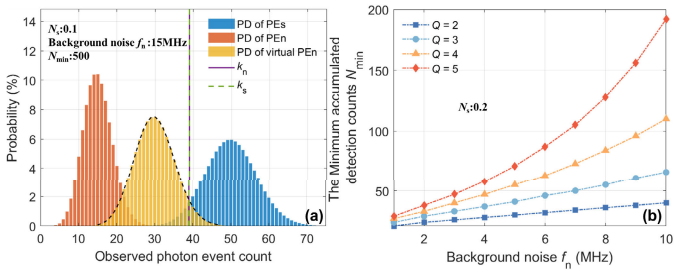


Fig. 10. (a) Schematic of the minimum cumulative detection count N_{min} under photon contrast constraints. The blue, orange, and yellow bars represent the PDFs of signal PE counts, noise PE counts, and equivalent noise PE counts, respectively. The solid purple line denotes k_n, while the green dashed line indicates k_s. Results indicate that achieving a photon contrast Q of 4.3 requires increasing the N_{min} to 500. (b) Relationship between N_{min} and Q, when the average signal photon number N_s is fixed at 0.2. The N_{min} grows rapidly as the required photon contrast increases.

$$\mu_{vn} = NP_{vn} = N(Q-1)P_n,$$
$$\sigma_{vn}^2 = NP_{vn}(1-P_{vn}) = N(Q-1)P_n - N(Q-1)^2 P_n^2. \quad (18)$$

This enables the derivation of the photon contrast constrained N_{min} by substituting P_{vn} into (11). Fig. 10 demonstrates that at 500 detections, k_s and k_n coincide precisely, indicating the critical point where signal and equivalent noise PE distributions become distinguishable with 90% accuracy. Meanwhile, the overlapping area between the signal and actual noise PE count reduces to a negligible level.

Fig. 10(b) presents the N_{min} required to achieve specified photon contrast Q at a fixed average signal photon number ($N_s = 0.2$), where square, circular, triangular, and diamond markers denote photon contrasts $Q = 2$, 3, 4 and 5, respectively. Theoretical results reveal a nonlinear growth in N_{min} with increasing Q, which means higher background noise rate amplifies the marginal detection count increments needed per unit gain in photon contrast.

C. Influence Analysis of Time Bin Width and Range Gate Length on Signal Discrimination

The theoretical model established in Section II assumes all signal PEs reside within a single time bin for analytical simplicity, enabling signal discrimination based on the PE counts within that time bin. However, practical LiDARs typically employ TCSPC system with timing resolutions far exceeding the laser pulsewidth to enhance ranging accuracy. When retaining the maximum-count bin as the discrimination criterion under these conditions, the required minimum detection count N_{min} is correlated with the time bin width.

As illustrated in Fig. 11(a), the curve depicts the relationship between the minimum cumulative detection count N_{min} and the time bin width τ. In the calculations, the center of the time bin with maximum PE count is aligned with that of the echo pulse, with a fixed background noise rate of 10 MHz and a laser pulsewidth (FWHM) of 1.16 ns. The circular, triangular, square, and diamond markers correspond

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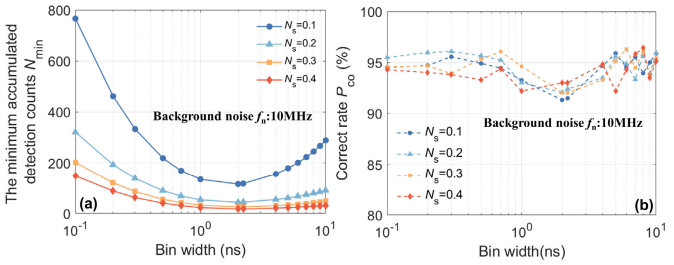


Fig. 11. (a) Theoretical relationship between the time bin width τ and the minimum cumulative detection count $N_{\min}$. (b) Discrimination accuracy P_{co} versus different time bin widths. The background noise rate and signal pulsewidth are fixed at 10 MHz and 1.16 ns. $N_{\min}$ reaches its minimum value when the bin width is identical with the signal pulsewidth.

to the theoretical calculation results for average signal photon number N_s = 0.1, 0.2, 0.3, and 0.4, respectively. Fig. 11(b) shows the discrimination accuracy P_{co} corresponding to the $N_{\min}$ in Fig. 11(a).

Fig. 11(a) reveals that as the τ increases, the $N_{\min}$ shows an initial decrease followed by an increase. $N_{\min}$ reaches the minimum value when τ approaches the signal pulsewidth ($\pm 2\sigma$s). When τ is less than the signal pulsewidth, the detection probability for signal PE (P_s) within the signal is significantly higher than that for noise PE (P_n). Consequently, the ratio P_s/P_n within a single time bin increases with τ. As demonstrated in Fig. 2(a), this results in a decrease in the $N_{\min}$ required for signal discrimination. When τ precisely encompasses the entire signal, the ratio P_s/P_n achieves its maximum, and $N_{\min}$ correspondingly reaches its minimum. Further increases in τ will not augment the count of signal PE, which means the ratio P_s/P_n begins to decline and leads to an increase in the $N_{\min}$. Regarding the signal discrimination accuracy P_{co}, when the time bin width is either lower or higher than the signal pulsewidth, it may lead to an underestimation of photon contrast. This further results in a larger detection count compared to the optimal $N_{\min}$ when τ exactly includes all signal PEs; thus, the discrimination accuracy P_{co} will remain higher than 90%. This analysis indicates that while a mismatch between the time bin width and the signal pulsewidth increases the $N_{\min}$, it does not degrade the signal discrimination accuracy.

In practice, the target position is typically unknown for a PC LiDAR system. The LiDAR system often employs a range gate spanning multiple time bins to cover potential target locations. Under this condition, correctly discriminating the target-reflected signal necessitates that the number of signal PEs within the target's time bin exceed the threshold k_s, as well as the number of noise PEs in other time bins within the range gate remains below the signal PE count. Therefore, as the length of the range gate increases, the discrimination accuracy decreases.

Fig. 12 presents a simulation for signal discrimination using a different range gate, under conditions of a N_s of 0.01 and a noise rate f_n of 5 MHz. The target-reflected signal is located at 60 ns, and the time bin width is 2 ns. The $N_{\min}$ calculated by (11) is 2423, with a discrimination threshold k_s of 42. The blue bars represent the PE count within each time bin. The time bin

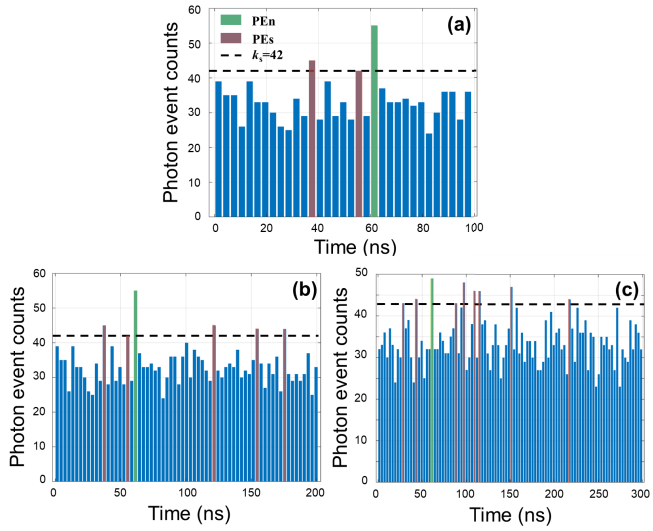


Fig. 12. Histograms of PEs accumulated by different range gate length. The blue bars denote the PE counts within each time bin. The time bin corresponding to the target is marked by green, while the time bins with noise PE counts exceeding k_s are shown in red. The discrimination threshold k_s = 42 is indicated by a black dashed line. From (a)–(c), the length of the range gate increases from 100 to 300 ns. The number of time bins marked with red grows as the length of the range gate increases, denoting a degraded false alarm rate.

containing the target-reflected signal is highlighted in green, while red bars denote time bins where the PE counts exceed k_s. As shown in Fig. 12(a), for a range gate of 100 ns, there are two time bins that exceed the threshold. When the length of the range gate increases to 200 and 300 ns, the number of false alarms rises significantly to five and eight, respectively. Thus, the increasing length of the range gate leads to a pronounced upward trend in the number of false alarms.

The false alarm rate of each time bin can be estimated by the probability that the number of noise PEs exceeds the number of signal PEs, as shown in the following equation:

$$P_{fa} = \frac{1}{2\pi\sqrt{\sigma_s\sigma_n}} \cdot \int_{k_s}^{+\infty}\int_{n_{PEs}}^{+\infty} \exp\left(-\frac{(n_{PEs}-\mu_s)^2}{2\sigma_s^2} - \frac{(n_{PEn}-\mu_n)^2}{2\sigma_n^2}\right) dn_{PEn} dn_{PEs}. \quad (19)$$

Assuming the length of the ranging gate is t_{rg} and the target time bin contains all signal PEs, the false alarm rate P_{Rfa} of the whole range gate can be derived as follows:

$$P_{Rfa} = 1 - (1 - P_{fa})^{t_{rg}/\tau} \quad (20)$$

where P_{fa} is calculated by (19). If a constraint on the false alarm probability is required, e.g., 5%, substituting (9) and (20) in the following equation yields the condition that the discrimination threshold k_s must satisfy

$$\operatorname{erf}\left(N_{\min}P_s - 1.65\sqrt{N_{\min}P_s(1-P_s)}\right) \geq 2\exp\left[\ln(1-P_{Rfa})\cdot\tau/t_{rg}\right]. \quad (21)$$

Based on the preceding analysis, we further investigate the false alarm rate P_{Rfa} under varying range gate lengths through theoretical calculations and Monte Carlo simulations.

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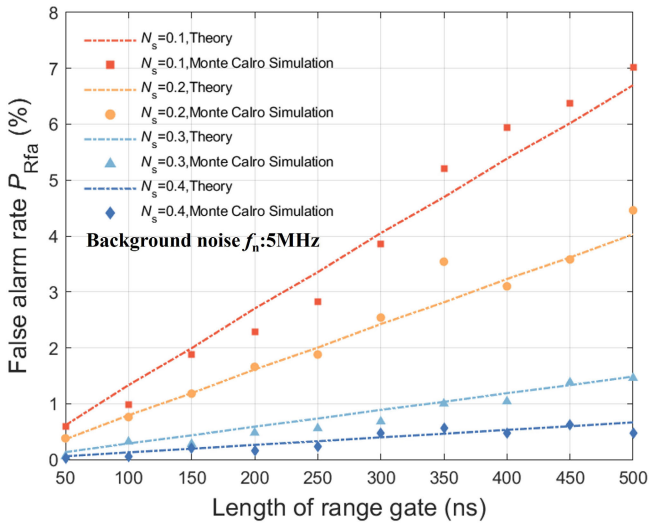


Fig. 13. False alarm rate P_{Rfa} of different range gate lengths. With the background noise rate fixed at 5 MHz, the curves and scatter points represent the theoretical calculation and Monte Carlo simulation results, respectively. The theoretical results show good consistency with the Monte Carlo simulations.

Fig. 13 illustrates the impact of range gate length on the false alarm rate P_{Rfa} under different photon contrast conditions. The curves represent theoretical calculation results with a fixed background noise rate of 5 MHz, while the circular, triangular, square, and diamond markers denote the Monte Carlo simulation results. The colors red, yellow, light blue, and dark blue correspond to average signal photon number N_{s} = 0.1, 0.2, 0.3, and 0.4, respectively. The abscissa represents the ranging gate length, while the ordinate represents the false alarm rate P_{Rfa}.

Both the theoretical calculations and the Monte Carlo simulation consistently confirm that under a given photon contrast, the false alarm rate P_{Rfa} increases with the length of the range gate. This trend is more pronounced under conditions of low photon contrast. Therefore, in remote sensing applications, if the signals to be identified originate from scattering such as clouds, aerosols, or water, the N_{min} required can be estimated using (19). Conversely, if the targets are hard targets that require precise localization, such as the land surface or shallow seabed, the influence of other time bins within the range gate should be considered. In this case, the requisite N_{min} can be obtained by solving for the numerical solution that satisfies (21).

D. Application in Scanning Imaging

Through the discussions in Sections IV-A–IV-C, this study extends and refines the applicability of the model, enhancing its adaptability to the practical requirements of LiDAR applications—particularly concerning the influence of photon contrast and range gate length on signal identification. In practice, LiDAR signal discrimination may need to satisfy multiple constraints simultaneously. Therefore, this section applies the proposed model to a LiDAR scanning imaging scenario. Based on the experimental system described in Section III-B, a 2-D scanning mirror (Thorlabs GSV002) is integrated into the transmitter to enable scanning capability. In conventional scanning imaging systems, the LiDAR system typically employs equal-interval scanning, where the imaging time or number of accumulated pulses is identical for each pixel. In contrast, the model proposed in this study allows the system to dynamically adjust the number of pulses accumulated per pixel based on the received signal and noise level. This approach is referred to as dynamic-interval scanning.

Fig. 14(a) shows the target used in the scanning imaging experiment. A 2-D scanning mirror is employed to control the laser beam for row-by-row scanning to reconstruct the image of the target. Fig. 14(b) presents the scanning image obtained under dark conditions (high photon contrast) using equal-interval scanning, whereas Fig. 14(c) shows the result acquired under low photon contrast conditions simulated by introducing a lamp as a noise source. Both images have a resolution of 100 × 100 pixels, with a fixed number of 5000 detections per pixel. The grayscale of each pixel corresponds to the normalized number of detected signal PEs reflected from the target. For signal discrimination, the time bin with the highest PE count in the cumulative histogram is identified as the target location. The number of PEs within the range of $\pm 2\sigma_s$ (the pulsewidth of the reflected signal) centered at this time bin is regarded as the signal PEs. The image in Fig. 14(c) is used as the ground truth. The structural similarity index measure (SSIM) is employed as a metric to evaluate the differences between the images obtained via equal-interval scanning and dynamic-interval scanning.

Fig. 14(d) demonstrates the number of accumulated pulses per pixel for dynamic-interval scanning, calculated using (11). Specifically, Fig. 14(d.1)–(d.3) corresponds to photon contrast constraints of Q = 2.0, 2.5, and 3.0, respectively. Fig. 14(e) presents three grayscale images obtained through dynamic-interval scanning using the theoretical counts from Fig. 14(d). Fig. 14(f) shows three corresponding grayscale images of the target acquired via equal-interval scanning. To ensure consistent total scanning imaging time, the number of accumulated pulses per pixel in equal-interval scanning is set to the average of the counts in Fig. 14(d); these average values are 289, 634, and 1210 from left to right, respectively.

A comparison of Fig. 14(d.1)–(d.3) reveals that as the required photon contrast constraint Q is increased from 2.0 to 3.0, the required detections per pixel rise significantly. Due to the low reflectivity of the dark numeric patterns “15” and “20” at the target center, the pixels corresponding to these regions require notably more detections than those in the surrounding brighter areas. This characteristic indicates that the dynamic-interval scanning method can allocate more detections to pixels with weaker signals under an identical total detection count, thereby enhancing the detection capability for weak signals. Comparing the results in Fig. 14(e.1) and 14(f.1), it can be observed that under the photon contrast constraint Q = 2, both scanning methods produce images with significant noise, yet they exhibit notable differences in detail. In the image acquired by dynamic-interval scanning, pixels with lower reflectivity receive more detections, resulting in reduced noise in the “15” and “20” patterns as well as the background. This leads to higher image contrast and sharper boundaries

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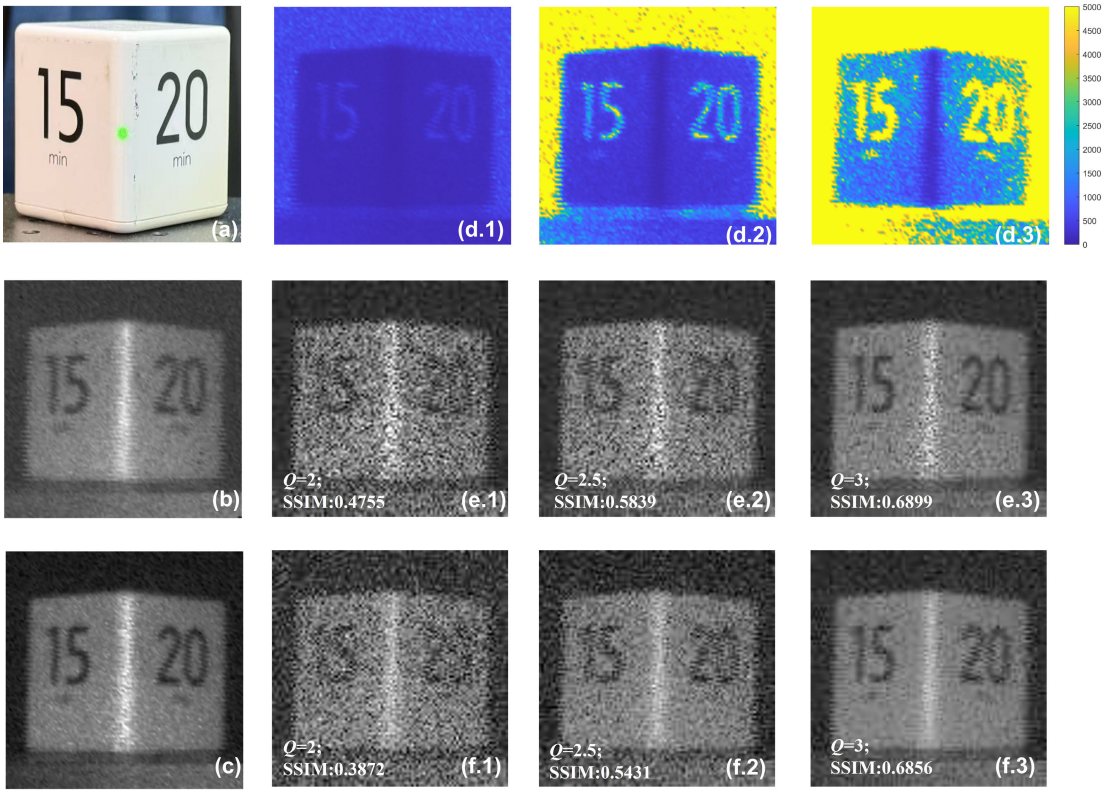


Fig. 14. Scanning imaging experimental results. (a) Target of the scanning imaging experiment. (b) Grayscale image acquired by the LiDAR system under high-SNR conditions, using equal-interval scanning method. The accumulated detection counts for each pixel is 5000. (c) Grayscale image acquired under low-SNR conditions, with identical scanning method and accumulated detection counts. (d.1)–(d.3) The accumulated detections of each pixel calculated by the proposed model under photon contrast constraints of Q = 2, 2.5, and 3, respectively. For regions where the upper limit of the photon contrast is lower than Q, e.g., the number "15" and "20," the minimum cumulative detection count $N_{\min}$ for these pixels are fixed at 5000. (e.1)–(e.3) Grayscale images reconstructed by the dynamic-interval scanning method under different photon contrast constraints. (f.1)–(f.3) Grayscale images acquired by equal-interval scanning method. The accumulated detection counts for each pixel are corresponding to the mean value of $N_{\min}$ in (d.1)–(d.3).

between the target and the background. As the photon contrast constraint is increased to Q = 2.5 and Q = 3, the SSIM between the dynamic-interval scanning results and the ground truth [Fig. 14(c)] rises to 0.6549 and 0.7522, respectively, validating the effectiveness of the proposed model. As the average number of accumulated detections increases, the SSIM between the equal-interval scanning results and the reference also rises to 0.5999 and 0.7250, respectively, but remains lower than that achieved by dynamic-interval scanning.

The experimental results demonstrate that the proposed model can significantly reduce the detection count required to discriminate the signal while satisfying the photon contrast constraint, thereby improving the detection efficiency of PC LiDAR systems. Particularly in scenarios where both the signal and noise vary dynamically, the proposed model can adaptively adjust the required number of pulses according to environmental and target conditions, benefiting data utilization efficiency. For instance, in the scenario of retrieving oceanic optical parameters using ICESat-2 ocean profiling data, the model can dynamically increase the accumulated detection data as water depth increases. This ensures the accuracy of signal photon identification, which is important for the stability of oceanic optical parameter retrieval.

V. Conclusion

This study establishes a theoretical model for determining the minimum detection count required for signal discrimination in PC LiDAR systems. First, a statistical model of signal and noise PE counts received in a time bin is constructed based on the binomial distribution. The concept of minimum detection count $N_{\min}$ is defined by the first occurrence of the 95th percentile of the noise PE distribution smaller than the 5th percentile of the signal PE distribution. By leveraging the normal distributions, an analytical expression for the minimum detection count $N_{\min}$ is derived, facilitating practical applications. Then the model is further extended to account for multichannel detector configurations and incorporates constraints on photon contrast, accounting for the effects of time bin width and range gate length on signal discrimination performance. Finally, the model is applied to a scanning imaging PC LiDAR, proposing a dynamic-interval scanning method that adjusts accumulated pulses per pixel based on real-time signal/noise levels.

The proposed model is validated through both Monte Carlo simulations and experiments. The simulation and experimental results confirm the accuracy and robustness of the proposed

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model. The results show close agreement between theoretical predictions and empirical data, i.e., the discrimination accuracy comparison exhibits mean absolute errors of 1.5% (simulation-theory) and 1.76% (experiment-theory), and maximum error less than 3%. The model balances detection efficiency and discrimination performance, particularly in challenging scenarios with high background noise or low photon contrast. Moreover, the application of the model in a scanning imaging system reveals that dynamic-interval scanning, which is guided by the derived model, enhances image quality under low photon contrast conditions. Compared with conventional equal-interval scanning, the proposed method reduces noise in low-reflectivity regions as well as improved SSIM without increasing total scan time. This result underscores the model's practical utility in improving LiDAR-based imaging where adaptive detection strategies are essential.

This study fills the gap in quantitative analysis for the minimum detection counts required by a PC LiDAR. We derive a closed-form expression for the minimum detection counts $N_{\min}$, which provides a unified calculation framework satisfying multiscenario constraints. These results may enhance both the theoretical understanding and practical applicability of PC LiDAR systems. The model presented in this study provides a quantitative a priori decision-making framework that may help support and inform key steps within the standard ICESat-2 (ATLAS) data processing chain. The proposed model could assist in optimizing PE classification in the ATL03 product by rapidly estimating the minimum cumulative pulse count from topographic constraints and system parameters, supporting more adaptive and physically consistent signal filtering. It may also provide some physical insight into the suitability of the 20 m geolocation segmentation currently used in the ATL08 product, suggesting that this segment length appears reasonable for reliable surface photon identification under moderate terrain conditions, while pointing to the potential for dynamic segment adjustment in steep-slope or low-reflectance regions. Additionally, the framework can be extended to inform oceanic applications, enabling the estimation of optimal along-track accumulation lengths for retrieving the diffuse attenuation coefficient K_d across varying water depths and photon contrast requirements. By providing theoretical support directly linked to ATL03, ATL08, and potential water-column product workflows, this approach may serve as a useful reference for enhancing the physical consistency and adaptive potential of operational ICESat-2 processing.

Future work may focus on real-time implementation of the model in field-deployable systems and its extension to more complex environments, such as underwater imaging or spaceborne PC LiDAR data processing.

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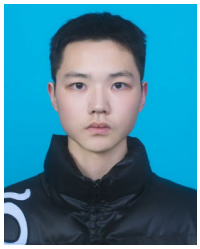
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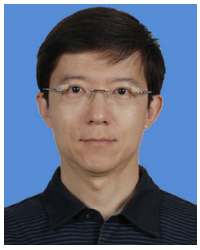
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